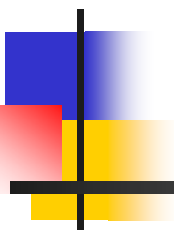
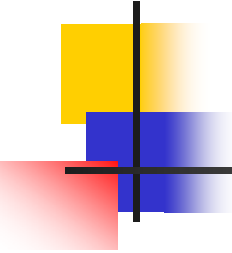


Introduction to Artificial Intelligence



- AI's Successes
- What is AI?
- Rationality
- A bit of History



Exciting Time for AI

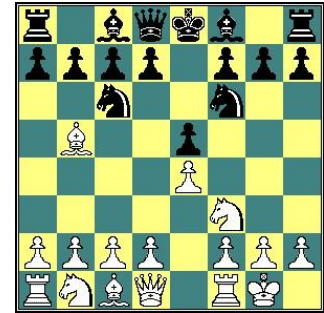
<http://aaai.org/AITopics/html/current.html>

- Expert Systems
 - Organic chemistry, Medicine, Geology, Configuring Computers, ...
 - User Help Systems [Microsoft]
- Web search engines
 - Google, Yahoo!, ...
- Natural Language Understanding
 - Eliza, Ask Jeeves, spell checkers, grammar checkers, ...
 - Real-time English-German speech translation!

AI Playing Games

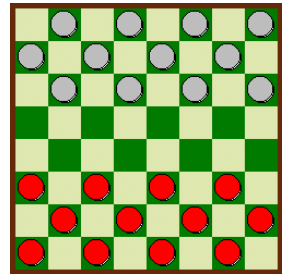
■ CHESS:

- *Deep Blue* BEAT Kasparov (May '97)
- Matches expert level performance
 - "intelligent & creative" play
- . . . thinks differently from human expert by examining 200million board-positions/move



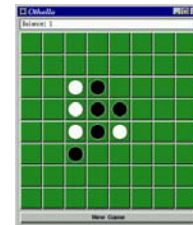
■ Checkers:

- *Chinook* is 1994 World champion CHECKERS Player!
- 2007: proven to be draw!



■ Othello

- Logistello beat Human Champion, 6-0 !



■ Backgammon

- TD-Gammon is competitive with World Champion



■ Poker

- Comparable with best humans! (2007)



Man vs Machine Poker

- Comparable with top human players (2007)
- Attracted international media attention.

“We won, not by a significant amount,
and the bots are closing in.”
– Phil Laak



“I really am happy it's over. I'm surprised we won ...
It's already so good it will be tough to beat in future.”
– Ali Eslami

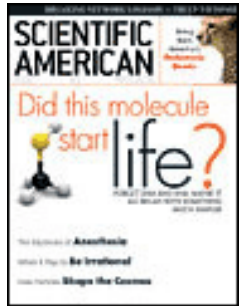


Playing Go

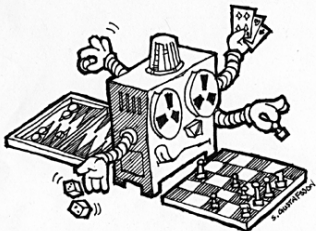


- First program to ever beat a human Go expert!
- Scientific American, June 2007

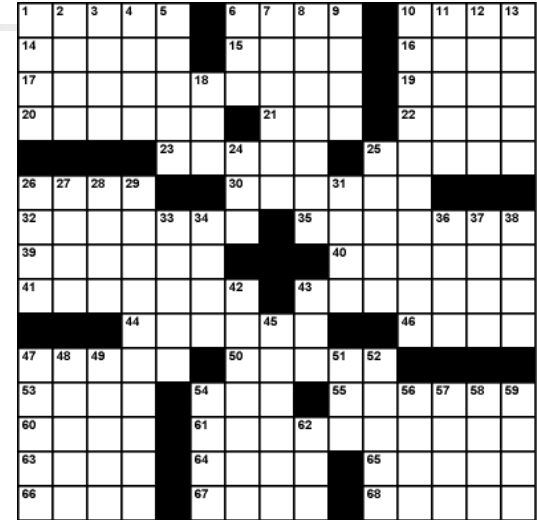
“UCT may prove useful for targeting advertisements on the Web, finding the best settings for an industrial plant or optimizing channel allocation in cellular systems.” – Scientific American



- UCT used to win 2007 General Game Playing (former UofA student)



Language "Understanding" + Problem Solving



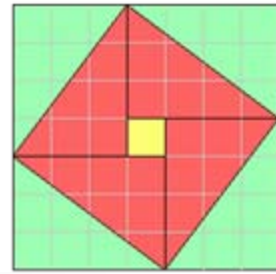
- PROVERB: solves cross-word problems using
 - many information sources
 - (geography, puns, . . .)
 - word fillers
 - past puzzles
 - . . .
- In NewYorkTimes puzzles
 - 95.3% words correct
 - 98.1% letters correct
 - ⇒ high-quality cruciverbalist!
- <http://www.oneacross.com>

Robotics

- Ping pong player
- Honda robot
- Automated car #1: 1996
 - "ALVINN": vision trained to steer car
 - "No hands across America"
 - over 2,805 miles, in control for 98% !
 - Different terrain, road conditions, weather,
- DARPA Challenge Oct 2005:
 - 132 miles
 - Completely autonomous
 - Stanley



Theorem Proving (Nov'96)



- First "creative" proof by computer
- Solved 60 year open problem.
 - Robbins' problem in finite algebra
 - <http://www.mcs.anl.gov/home/mccune/ar/robbins>
- Qualitative difference from previous brute-force results
- . . . unlike [Appel/Haken] computer proof of "4 color" theorem



Decision theory and Statistical User-models

- Lymph node pathology
Explain response. . .
- Microsoft Office '97 + Answer Wizard
Diagnosis/reasoning using Bayesian
models
<http://www.research.microsoft.com/research/dtg/>
- Restricted natural language parsing



Logistics Planning

- Gulf War, 1991
 - 50,000 people, cargo, vehicles
 - various start and designation points
- "DART" (AI System)
 - produced plan in hours;
 - by hand: many weeks
- DARPA:
"this single application paid back 30 years of funding" !



Autonomous Intelligent Systems (NASA)



- Next generation spacecrafts
"DS1" (launched mid98)
- Control space-craft and carry out objectives, while maintaining internal systems, ... over decades.
- Fast real-time, on-line performance
- Automatic planning and execution model
- General deduction from first principles
synthesize timely responses to anomalous & unexpected situations

<http://ic-www.arc.nasa.gov/ic/projects/mba/>

Machine Learning Successes: Computer learns...

AMERICAN
EXPRESS



- to find ideal customers

Credit Card approval (AMEX)

- Humans $\approx 50\%$; ML is $>70\%$!

- to find best person for job

Telephone Technician Dispatch [Danyluk/Provost/Carr 02]

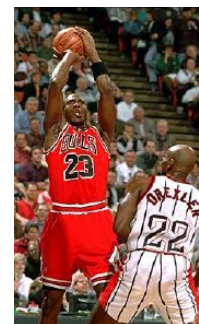
- BellAtlantic used ML to learn rules to decide which technician to dispatch
- Saved \$10+ million/year

- to predict purchasing patterns

- Victoria Secret (stocking)

- to help win games

- NBA (scouting)

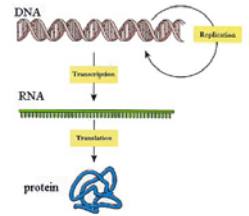


- to catalogue celestial objects [Fayyad et al. 93]

- Discovered 22 new quasars
- $>92\%$ accurate, over tetrabytes

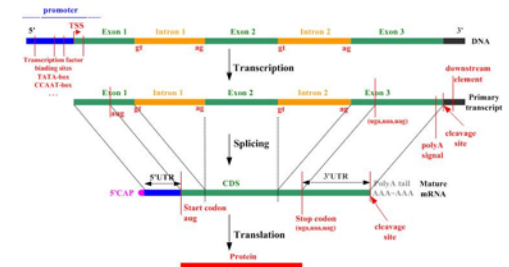


ML#2: Sequential Analysis

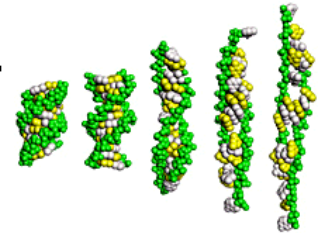


■ BioInformatics 1: identifying genes

- Glimmer [Delcher et al, 95]
- identifies 97+% of genes, automatically!



■ BioInformatics 2: Predicting protein function, ...



■ Recognizing Handwriting

Also, Brakes 1-1-0
 Bought to knife - 0-0-0
 for Skating, 4-1-0
 for from Black - 0-0-0
 of the play - 0-9-0
 (handwritten text)

US postal service zip code reader

for my self - 3-2-02-0
 had left in my garage 3-1-0
 (handwritten text)

■ Recognizing Spoken Words

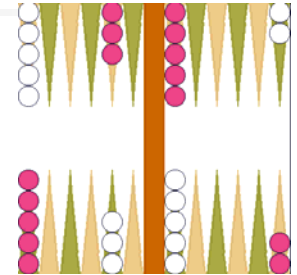
- “How to wreck a nice beach”

Ticketmaster, Speechworks, Bell, Verbmobile [translation]

ML#3: Control

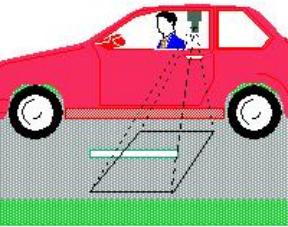
- **TD-Gammon** (Tesauro 1993; 1995)

- World-champion level play by **learning** ...
- by playing millions of games against itself!



- **Drive autonomous vehicles** (Thrun 2005)

- DARPA Grand Challenge



- **Printing Press Control** (Evans/Fisher 1992)

- Control rotogravure printer, prevent grooves, ... specific to each plant
- More complete than human experts
- Used for 10+ years, reduced problems from 538/year to 26/year!



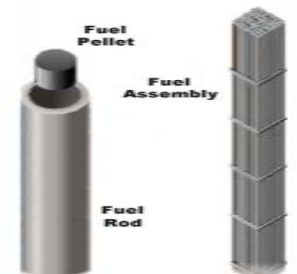
- **Oil refinery**

- Separate oil from gas
- ... in 10 minutes (human experts require 1+ days)



- **Manufacture nuclear fuel pellets** (Leech, 86)

- Saves Westinghouse >\$10M / year



- **Adaptive** agents / user-interfaces

Industry Interest in ML



Microsoft®

"If you invent a breakthrough in artificial intelligence, so machines can learn, that is worth 10 Microsofts."
– Bill Gates, New York Times, 1/03/2004



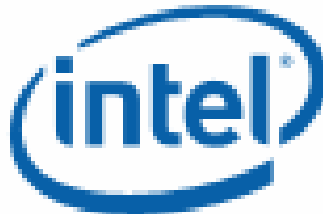
Google™



YAHOO!®



IBM



intel®



hp®
invent



Sun
microsystems



WAL★MART®



SIEMENS



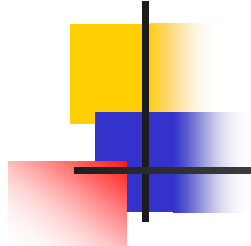
**AMERICAN
EXPRESS**



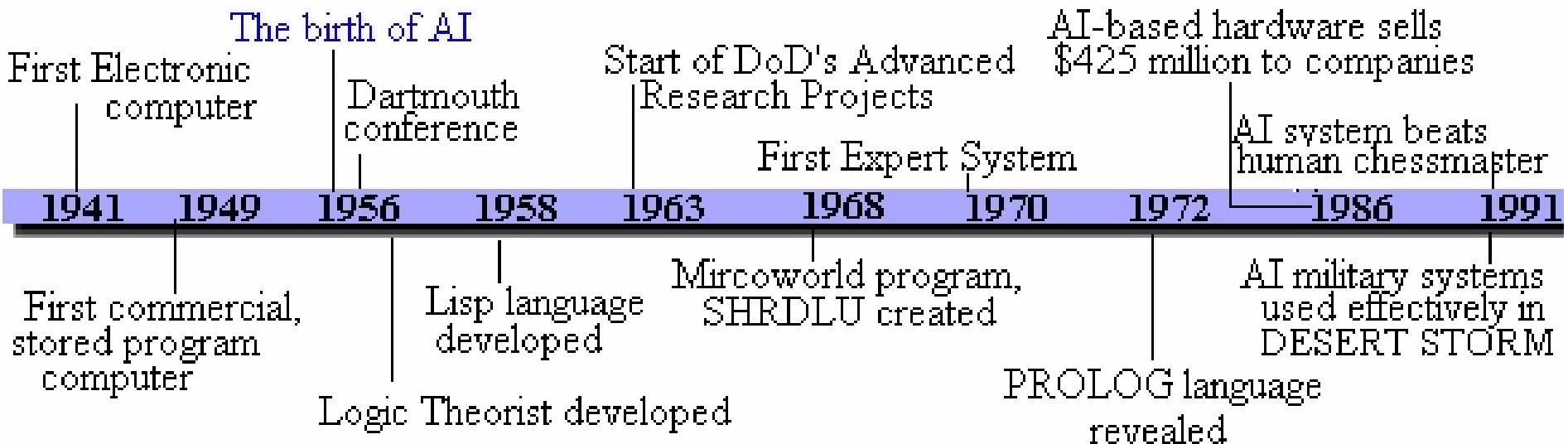
amazon.com™



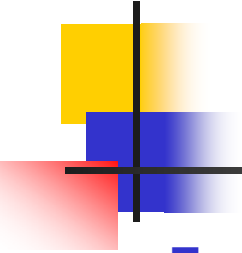
BOEING



AI History



- Foundations...
- Technology push
- Artificial domains
- Real applications



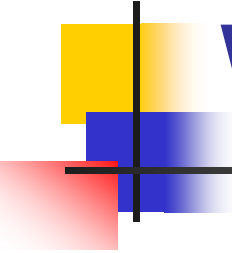
Trends Since 90's

- Relying less on logic;
more on *probability theory and statistics*
- More emphasis on *objective performance evaluation*
- Intelligent (rational) Agents
- Accomplishments in
 - Game playing: Deep Blue, Chinook, Logistello, ...
 - Space Probe
 - Biological sequence analysis
 - OCR
 - Consumer electronics



Questions ... for audience!

- What is Artificial Intelligence?
 - ? artificial HUMAN intelligence ?
 - ? intelligent ARTIFACTS ?
 - ? applied EPISTEMOLOGY ?
 - Science? Math? Psychology? Philosophy? . . .
- What does AI involve?
 - ... not involve?
 - ? Reasoning/ProblemSolving/"Thinking" ?
 - ? Game playing ?
 - ? (Intelligent) Agents ?
 - ? Image Understanding ?
 - ? Robotics ?
 - ? Natural Language / Communication ?
 - ? Learning ?
- Can it succeed?
- Why should we care if it does?
 - ? Previous successes ?
 - ? Anticipated successes ?
- What are open problems?



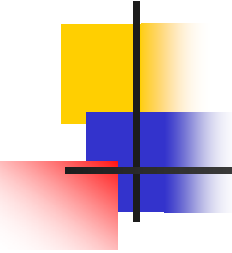
What is Artificial Intelligence?

AI studies systems (eg, computer programs)
that ...

1. Act like humans	4. Act rationally
2. Think like humans	3. Think rationally

?

?



1. Act Like Humans

1. Act like humans	4. Act rationally
2. Think like humans	3. Think rationally

- AI = ...

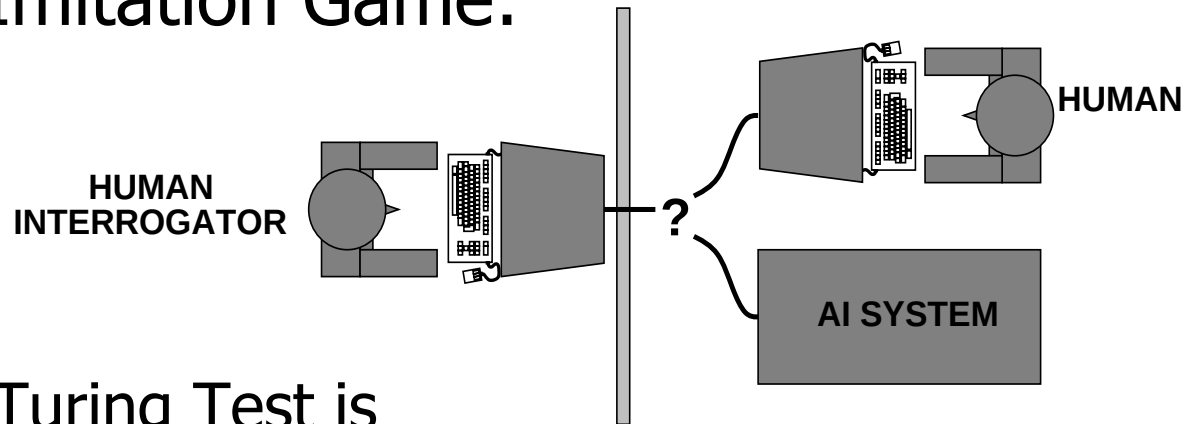
creating machines that perform
“intelligent functions”
... only by intelligent humans

- Methodology:
Find an intellectual task
and
make a computer do it

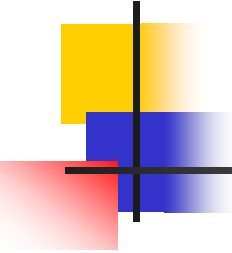
- Prove a theorem
- Play chess
- Plan a surgical operation
- Diagnose a disease
- Navigate in a building

Turing Test

- *NOT:* "Can machines think"
- *Instead:* "Can machines behave intelligently?"
- Operational test of intelligent (human like) behavior: Imitation Game:



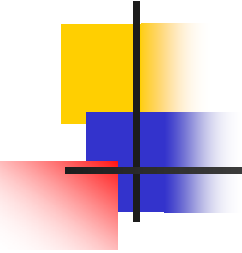
- Problems: Turing Test is
 - not reproducible,
 - not constructive,
 - not amenable to mathematical analysis



2. Think Like Humans

1. Act like humans	4. Act rationally
2. Think like humans	3. Think rationally

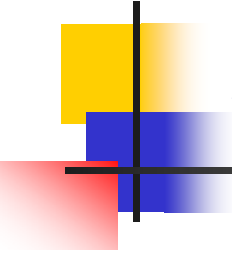
- 1960s “cognitive revolution”:
information-processing psychology replaced *behaviorism*
- Requires scientific theory of internal activities of the brain
 - What level of abstraction?
“Knowledge” or “circuits”?
 - How to validate? Requires
 1. Predicting and testing behavior of human subjects (top-down) vs
 2. Direct identification from neurological data (bottom-up)
- Both approaches
(≈ Cognitive Science and Cognitive Neuroscience)
are now distinct from “core AI”



Examples

- Garden-Path Sentence:
 - The horse raced past the barn fell.
- Center-embedding:
 - The cat that the dog that the mouse that the elephant admired bit chased died.
 - The elephant admired the mouse that bit the dog that chased the cat that died.
- Multiplication: $3242342 * 2342342$

Do we want to duplicate human imperfections?



Artificial Human Intelligence??

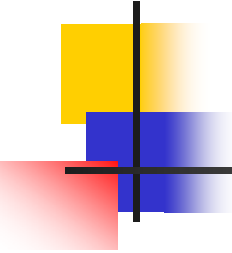
- How to build good FLYING machine
 - [?] Synthesize birds?
 - [?] Understand aerodynamic principles
- Best FLYING Machine = Airplane
≠ Improved Bird
- How to build good THINKING machine
 - [?] Synthesize human brains?
 - [?] Understand reasoning principles
- Best THINKING Machine ?=? Improved Human Brain
[. . . DeepBlue . . .]

3. Think Rationally:

Laws of Thought

1. Act like humans	4. Act rationally
2. Think like humans	3. Think rationally

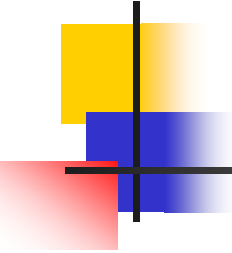
- Normative (or prescriptive), not descriptive
- Aristotle:
 - What are correct arguments/thought processes?
 - Several Greek schools developed forms of *logic*: notation and rules of derivation for thoughts.
- Problems:
 - Not all intelligent behavior is mediated by logical deliberation
 - What is the purpose of thinking?
What thoughts ***should*** I have?



4. Act Rationally

1. Act like humans	4. Act rationally
2. Think like humans	3. Think rationally

- Rational behavior: *doing the right thing*
- “The right thing” $\equiv$
action that is expected to
maximize goal achievement,
given the available information
- Doesn't necessarily involve thinking
(eg, blinking reflex)...
but
thinking should be in service of
rational action



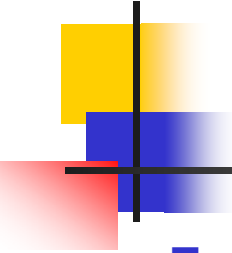
Rational Agents

- AI $\equiv$ Study of design of rational agents
- agent = thing that acts in environment
- Rational agent = agent that acts rationally:
 - actions are appropriate for goals and circumstances
 - makes appropriate choices given
 - perceptual limitations and finite computation
 - (flexible to changing environments and goals; learns from experience)
- Details later...

Goals of Artificial Intelligence

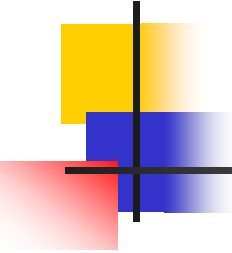
- Scientific goal:
 - understand principles that make rational (intelligent) behavior possible, in natural or artificial systems.
- Engineering goal:
 - specify methods for design of useful, intelligent artifacts.
- ~~■ Psychological goal:
 - ~~■ understanding/modeling people~~
 - ~~■ cognitive science (not this course)~~~~





Goals of This Course

- Introduce key methods & techniques from AI
 - searching
 - reasoning and decision making
 - logical
 - decision theoretic (probabilistic)
 - ? learning (covered in detail in CMPUT466)
 - ? language processing/understanding,
 - ? vision/robotics
 - . . .
- Understand applicability and limitations of these methods



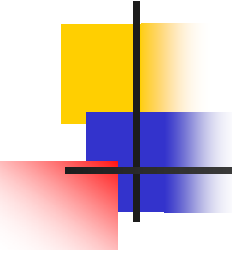
Approach of Course

- Our approach:
 - Characterize Environments
 - Identify rational agent that is most effective for each environment
- Study increasingly complicated agent architectures requiring
 - increasingly sophisticated representations,
 - increasingly powerful reasoning strategies

To help set expectations...

State of the Art:

- Play a decent game of table tennis
- Drive safely along a curving mountain road
- 😊 ■ Drive safely along White Avenue
- Buy a week's worth of groceries on the web
- 😊 ■ Buy a week's worth of groceries at Safeway
- Play a decent game of bridge
- 😊 ■ Discover and prove a new mathematical theorem
- 😊 ■ Design and execute a research program in molecular biology
- 😊 ■ Write an intentionally funny story
- Give competent legal advice in a specialized area of law
- 😊 ■ Translate spoken English into spoken German in real time
- 😊 ■ Converse successfully with another person for an hour
- 😊 ■ Perform a complex surgical operation
- 😊 ■ Unload any dishwasher and put everything away



Summary

- AI Successes
- What is AI?
- Rationality
- A bit of History
- Expectations...